Lab 1 Report: Introduction to AI & Paradigms

CAI601400 - Artificial Intelligence
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1. Problem and Assumptions

A university wants an AI helpdesk/screening prototype. The team must decide what makes a system "AI", what evidence shows intelligent behaviour, and what a rule-based and a data-driven approach can and cannot do. The case study is a scholarship screening prototype that flags incomplete applications.

- Review is a flag for a human reviewer, not an award decision.
- All thresholds (score >= 70, GPA >= 2.5, letter required) are synthetic lab assumptions.
- Probabilities in lab1_compare.py are invented demonstration inputs. No model is trained, so the comparison shows behaviour, not accuracy.

2. Task 1: AI Milestone Timeline

Year/Period	Event	Significance	Source
1950	Turing publishes "Computing Machinery and Intelligence" (imitation game)	Proposed a behavioural, task-based way to discuss machine intelligence.	Turing, A. M. (1950). Mind, 59(236), 433-460.
1956	Dartmouth Summer Research Project on AI	Founding event that named the field and set its early symbolic research agenda.	McCarthy et al. (1955), Dartmouth proposal; Russell & Norvig, AIMA, Ch. 1.
1974-1980 (first AI winter)	Lighthill Report (1973) and funding cuts in UK/US	Overpromised results led to reduced funding and expectations.	Lighthill, J. (1973), Artificial Intelligence: A General Survey; Russell & Norvig, AIMA, Ch. 1.
1980s	Expert systems boom (e.g. XCON at DEC), later followed by a second winter (late 1980s-1990s)	Showed value of explicit rules but also their brittleness and maintenance cost.	Russell & Norvig, AIMA, Ch. 1.
2012	AlexNet wins ImageNet using GPUs	Showed that large datasets + GPU compute + learning algorithms drive a step change in performance.	Krizhevsky, Sutskever & Hinton (2012), NeurIPS 25.

Note: dates and sources are from standard references; please verify each citation against your own copy before submission.

3. Implementation Choices

Two scripts were written in plain Python with no external packages.

- lab1_compare.py: a symbolic function (score >= 70 and documents complete) and a threshold function (probability >= 0.70) run on four cases.
- lab1_rules.py: the rule system extended with two named conditions, min_gpa_met (GPA >= 2.5) and recommendation_letter_present. Each decision returns a reason string so the logic stays traceable.

4. Results

4.1 Disagreement table (symbolic vs threshold)

Case	Score / Complete / Prob.	Symbolic	Threshold	Disagree ?	Explaining condition
complete_high	82 / True / 0.81	Review	Review	False	Both conditions pass
complete_low	68 / True / 0.74	Hold	Review	True	score >= 70 fails; prob 0.74 >= 0.70
incomplete_high	91 / False / 0.88	Hold	Review	True	documents_complete fails; threshold ignores it
boundary	70 / True / 0.70	Review	Review	False	>= includes the boundary in both

Captured output:

```
case | symbolic | threshold | disagreement
complete_high | Review | Review | False
complete_low | Hold | Review | True
incomplete_high | Hold | Review | True
boundary | Review | Review | False
```

Isolation test (Section 1.6): score=69 gives Hold, probability=0.69 gives Hold, and score=70 with incomplete documents gives Hold. This confirms the boundary is inclusive for both rules and that completeness alone can block a high score.

4.2 Task 2: four core rule tests

Case	Input	Expected	Actual	Reason
pass_all	score=82, docs=True, gpa=3.2, letter=True	Review	Review	all conditions satisfied
fail_low_gpa	score=85, docs=True, gpa=2.1, letter=True	Hold	Hold	min_gpa_met failed (2.1 < 2.5)
fail_no_letter	score=88, docs=True, gpa=3.5, letter=False	Hold	Hold	recommendation_letter_present failed
boundary_exact	score=70, docs=True, gpa=2.5, letter=True	Review	Review	all conditions satisfied

Test selection: one all-pass case, one failure for each new condition (the other conditions held satisfied so the cause is isolated), and one exact-boundary case (score 70, GPA 2.5) to check that >= is implemented correctly. All four matched expectations.

5. Failure / Edge-Case Analysis

- Input: incomplete_high (score 91, documents incomplete, probability 0.88).
- Observed: the symbolic rule returns Hold; the threshold rule returns Review.
- Likely cause: the probability value carries no explicit completeness condition, and how the score was produced is not visible.
- Improvement: add a hard completeness check before any score-based step, and log the reason for every flag. This disagreement is not evidence that either approach is better, since there is no ground-truth data.

6. Task 3: Conversational Test vs General Intelligence

A conversational (Turing-style) test observes whether a judge can distinguish a machine's replies from a human's in dialogue. It does not establish reasoning reliability, correct rule-following, understanding, or general intelligence across tasks. Fluency can hide errors.

Example of fluent but incorrect reasoning: an assistant is asked whether an application with a score of 91 but a missing transcript should be flagged. It replies confidently that the strong score justifies Review and politely explains why, overlooking the documented completeness rule. The explanation sounds convincing yet violates the policy.

Additional test of reliable task performance: run the assistant on a fixed set of labelled applications (including incomplete ones and exact-boundary cases) and measure how often its flag matches the documented rule, repeating runs to check consistency.

7. Limitations

- Synthetic thresholds and invented probabilities; results do not generalise.
- Only four cases per experiment; no ground-truth labels, so no accuracy claim is possible.
- Real use would need validated data, clear policy ownership and human review.

8. Reflection

- What I learned: rules are traceable but brittle, while score-based methods hide how the number was produced; an AI winter followed overpromising.
- How I will apply it: keep hard policy conditions as explicit rules and use learned scores only as one input, with human review.
- What surprised me: a high score or probability can still hide a policy violation such as a missing document.
- What I would do differently: add more edge cases and test an actual trained classifier with labelled data.

9. Responsible Practice

Claims are tied to observed outputs, not marketing language. Assumptions are recorded, no private data or credentials are used, and the system only flags cases for human decision.

10. References

- Turing, A. M. (1950). Computing Machinery and Intelligence. Mind, 59(236), 433-460.
- McCarthy, J., Minsky, M., Rochester, N., Shannon, C. (1955). A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence.
- Lighthill, J. (1973). Artificial Intelligence: A General Survey. Science Research Council.
- Russell, S., Norvig, P. Artificial Intelligence: A Modern Approach, Ch. 1.
- Krizhevsky, A., Sutskever, I., Hinton, G. (2012). ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 25.